



PROFESSIONAL CERTIFICATE · INSTRUCTOR-LED

Agentic AI Bootcamp

A 76-hour, build-first program that takes engineers from a bare-metal agent loop to a deployed, evaluated, guard-railed multi-agent system — self-hosted and on the managed cloud (Microsoft Foundry).

76 hours

16 modules + capstone

~68% hands-on

Python · LangGraph · CrewAI · MCP · Skills · A2A · Foundry

7

Phases

16

Modules

~52h

Labs & build time

1

Production capstone

PROGRAM AT A GLANCE

Designed for graduate students and working engineers with basic Python. Framework-agnostic principles first, then implementation — every module ends with a working artifact, not a slide.

Pedagogy: **Build → Test → Improve.**

AUDIENCE

Graduate students & engineers with basic Python proficiency

FORMAT

Instructor-led workshop · cohort of ~20 learners

CORE STACK

Python · LangGraph · CrewAI · MCP · A2A · Microsoft Foundry · GitHub Copilot

OUTCOME

Job-ready: design, build, evaluate, and deploy agentic AI systems — self-hosted and on the cloud (Microsoft Foundry)

EACH MODULE SHIPS

Learner Handout · Lab Worksheet · starter + solution code (Phase 7: branded notebooks + weekly mini-projects)

WHAT YOU'LL BE ABLE TO DO

- Tell a single LLM call apart from a workflow and a true autonomous agent — and build each.
- Implement ReAct, planning, and reflection patterns with structured, parse-safe outputs.
- Connect agents to live tools through the Model Context Protocol (MCP).
- Give agents persistent long-term memory that survives restarts and cites sources.
- Orchestrate multi-agent crews and let frameworks interoperate via A2A.
- Evaluate, trace, guard-rail, and deploy agents under real cost and latency budgets.
- Rebuild agents on the managed cloud — Microsoft Foundry Agent Service, grounding, content safety, and evaluation.

CURRICULUM · 7 PHASES

Foundations & Mental Models

PHASE 1

9h

M0-M2

From “what is an LLM” and “what is an agent” to a reproducible engineering setup and a framework-agnostic mental model.

M0

LLM & Reasoning Primer

~1h

Pre-work primer: understand the LLM you're about to augment — a token-in/token-out function, its context window, temperature, and parametric memory — and why reasoning LLMs (“thinking tokens”) improve planning and tool selection. Hands-on micro-lab feels statelessness, temperature, and reasoning vs plain models.

DELIVERABLE LLM/reasoning notes + micro-lab

Concept / Lab: 0.7 / 0.3h

M1

What Agentic AI Really Is

4h

Build a bare-metal agent loop in pure Python (no frameworks) and a one-page “Is this an agent?” decision note. Distinguish a single LLM call vs a workflow vs an autonomous agent, and run the perceive → reason → act → observe loop with a step budget and cost counter.

DELIVERABLE Bare-metal agent loop + decision note

Concept / Lab: 1.5 / 2.0h

M2

Architectures & the 2026 Stack

4h

Stand up a reproducible repo scaffold (venv, lockfile, git-ignored secrets) and the same “hello agent” in both LangGraph and CrewAI. Draw the agentic reference architecture and correctly place MCP (agent ↔ tools) vs A2A (agent ↔ agent).

DELIVERABLE Repo scaffold + dual-framework hello agent

Concept / Lab: 2.0 / 1.5h

Core Agent Design Patterns

PHASE 2

10h

M3-M4

The two patterns that make agents reliable: structured ReAct + planning, and reflective self-critique.

M3

ReAct, Planning & Tool-Use

5h

Build a graph-based ReAct + Planning agent in LangGraph that decomposes multi-step questions, uses structured outputs (no fragile JSON parsing), and replans — with a visualized state graph and a failure-mode note.

DELIVERABLE

Graph-based ReAct + Planning agent

Concept / Lab: 1.5 / 3.0h

M4

Reflection, Self-Critique & Robustness

5h

Build a Self-Improving Research Agent (critic-actor loop) that drafts, critiques itself against a rubric, and revises — bounded by iteration, cost, and “good-enough” stops — plus hardening against loops, hallucinated tools, and oscillation.

DELIVERABLE

Self-Improving Research Agent

Concept / Lab: 1.5 / 3.0h

Tools, Skills, Integrations & Memory

PHASE 3

14h

Connect agents to the world through MCP, package reusable Agent Skills, and give them short- and long-term memory that persists.

M5-M6

M5

Tools, APIs & MCP

5h

Build a custom MCP server (typed tools, validation, timeouts) and a LangGraph agent that consumes it as an MCP client, calling live tools through the protocol — with documented tool schemas.

DELIVERABLE

Custom MCP server + MCP-client agent

Concept / Lab: 1.5 / 3.0h

M5B

Agent Skills

4h

Package a recurring task as a reusable Agent Skill — a self-contained folder of instructions + optional scripts/resources the agent loads on demand via progressive disclosure. Distinguish a skill from a tool, an MCP server, and fine-tuning, then load a skill into your M5 agent so it fires only on matching tasks.

DELIVERABLE

Reusable skill loaded into your agent

Concept / Lab: 1.0 / 3.0h

M6

Memory Systems (Short & Long-Term)

5h

Build the Personal Knowledge Assistant — short-term summarization plus persistent long-term vector memory that recalls preferences across a restart and cites sources, with a documented memory policy.

DELIVERABLE

Personal Knowledge Assistant

Concept / Lab: 1.5 / 3.0h

Multi-Agent Systems & Protocols

PHASE 4

12h

Compose specialists into crews, orchestrate them safely, and let frameworks talk to each other via A2A.

M7-M9

M7

Multi-Agent Foundations (CrewAI)

4h

Build a 3-role CrewAI crew (Researcher → Analyst → Writer) that turns one brief into a structured report, with per-role tools and a single-vs-multi-agent reflection.

DELIVERABLE 3-role research crew

Concept / Lab: 1.0 / 2.5h

M8

Orchestration Patterns & Control Flow

4h

Build a LangGraph supervisor/router that routes tasks to specialists, fans out work in parallel, aggregates results, and always terminates within global budgets — with a topology visualization.

DELIVERABLE Supervisor / parallel router

Concept / Lab: 1.5 / 2.0h

M9

Agent-to-Agent Communication (A2A)

4h

Build the Cross-Framework Agent Network — a supervisor delegates across a CrewAI crew and a LangGraph agent via A2A, each using MCP tools — with an architecture diagram labeling both layers.

DELIVERABLE Cross-framework agent network

Concept / Lab: 1.5 / 2.0h

Production, Evaluation & Safety

PHASE 5

10h

Measure what you built, then harden it: tracing, evals, guardrails, approval gates, and cost control.

M10–M11

M10

Evaluation & Observability

5h

Instrument the Phase-4 system with tracing, build a ~15-task eval suite producing a scorecard (success rate, cost, p95 latency), and ship before/after fixes of the two worst cases diagnosed from traces.

DELIVERABLE

Eval suite + traced scorecard

Concept / Lab: 1.5 / 3.0h

M11

Guardrails, Safety & Cost/Latency

5h

Build the Production-Hardened Agent — input/output guardrails, an injection red-team (blocked), a human-in-the-loop approval gate, plus model routing + caching with a measured cost/latency improvement and a one-page reliability report.

DELIVERABLE

Production-Hardened Agent

Concept / Lab: 1.5 / 3.0h

Deployment & Advanced Patterns

PHASE 6

4h

Ship the self-hosted version: a deployed, callable agent service with health checks, tracing, and advanced patterns.

M12

M12

Deployment & Advanced Patterns

4h

Ship a deployed, callable agent service (FastAPI + Docker) with health and trace verification, plus advanced patterns (async/long-running tasks, streaming, externalized state) and a deployment checklist.

DELIVERABLE

Deployed agent service

Concept / Lab: 1.0 / 3.0h

Agentic AI in the Cloud — Microsoft Foundry

PHASE 7

Rebuild your agents on Microsoft Foundry — the managed cloud twin: model deployments, hosted Agent Service, grounding, content safety, tracing, and evaluation.

12h

M13–M15

M13

Microsoft Foundry & Cloud Models

4h

The managed cloud twin of M1–M2: deploy a model in a Microsoft Foundry project and call it from your M1 harness — no keys in code (Entra ID auth), with model catalog, deployments, and cost/governance basics.

DELIVERABLE

Model deployed in Foundry, called from Python

Concept / Lab: 1.5 / 2.5h

M14

Cloud Agents & Knowledge

4h

The Foundry Agent Service is the managed version of the M1 loop; grounding is managed M6 memory. Stand up a hosted agent, attach knowledge (Foundry IQ / Azure AI Search), and return grounded, cited answers — optionally wiring an MCP tool.

DELIVERABLE

Grounded Foundry agent (managed RAG)

Concept / Lab: 1.5 / 2.5h

M15

Orchestration, Safety & Observability in Foundry

4h

The managed twins of M8/M10/M11: add Content Safety + Prompt Shields, connect tracing to Application Insights, and run a Foundry evaluation to produce a scorecard — the Phase-7 Cloud mini-project.

DELIVERABLE

Guardrailed, traced, evaluated cloud agent

Concept / Lab: 1.5 / 2.5h

CAPSTONE • CODE OR CLOUD TRACK

Production-Grade Agentic System

A complete, deployed, evaluated, guard-railed multi-agent system that integrates the whole program — delivered with a source repo, architecture diagram, evaluation scorecard, reliability notes, and a **10-minute live demo + Q&A**. Teams choose a **Code track** (self-hosted) or a **Cloud track** (Microsoft Foundry).

ASSESSMENT & DELIVERY

COMPONENT	WEIGHT	CADENCE OPTION	LAYOUT
Capstone project	30%	Intensive	6 hrs/day × 12 days
Mini-projects (6 · incl. Cloud)	35%	Part-time	2 × 4h sessions/week (~9 weeks)
Practical evaluations (2)	15%	Weekend	Fri (remote) + Sat (in-person), 4h each · 19 sessions (~11 weeks)
Phase quizzes (7)	10%		
Participation & labs	10%		Hands-on ratio ~ 68% · ~24h concept / ~52h labs & build.

Pass bar: **≥70% overall** and a working capstone meeting all mandatory requirements.



Delivered in partnership with **JHF** & **COMCEC** · Lead trainer **Alaaldin Ahmed**